

STK435 IC-Based STEREO AMPLIFIER

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This stereo AF amplifier uses the STK435 IC, a highly popular choice due to its simplicity. The STK435, with its 15-pin configuration, requires only a few external components, such as electrolytic and ceramic capacitors and resistors. With a 12V DC power supply, the amplifier delivers approximately 7 watts to an 8-ohm

speaker, offering excellent audio quality. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the STK435-based stereo amplifier is shown in Fig. 2. The design includes the STK IC, resistors, and capacitors, with two audio inputs and corresponding amplified outputs for the right and left channels.

Start by completing the connections for the first amplifier section of IC1, referred to as IC1A. Pin 1 connects to two resistors (R2 and R3) in series, followed by a 100-ohm resistor (R5) leading to the +12V

supply. A 10 μ F, 25V electrolytic capacitor (C3) is connected between the junction of R2 and R3 and the ground. A 0.47 μ F, 25V electrolytic capacitor (C2) connects to pin 1

PARTS LIST

Semiconductor:

IC1 (IC1A, IC1B) - 15-pin STK435 stereo amplifier IC

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R15 - 1-kilo-ohm
R2, R12, R4, R14 - 220-kilo-ohm
R3, R13 - 390-kilo-ohm
R5 - 100-ohm
R6, R11 - 120-ohm
R7, R9 - 4.7-ohm
R8, R10 - 12-kilo-ohm

Capacitors:

C1, C13 - 470pF ceramic disk
C2, C12 - 0.47 μ F, 25V electrolytic
C3, C11 - 10 μ F, 25V electrolytic
C4, C9 - 220 μ F, 25V electrolytic
C5, C10 - 47 μ F, 25V electrolytic
C6, C7 - 1000 μ F, 25V electrolytic
C8, C14 - 0.1 μ F ceramic disk
C15 - 100 μ F, 25V electrolytic

Miscellaneous:

CON1-CON3 - 2-pin connectors
LS1, LS2 - 8-ohm, 7-watt speakers with 2-pin connectors
- 7cm x 9cm Veroboard
- 12V battery
- Heat sink for IC STK435



Fig. 1: Author's prototype

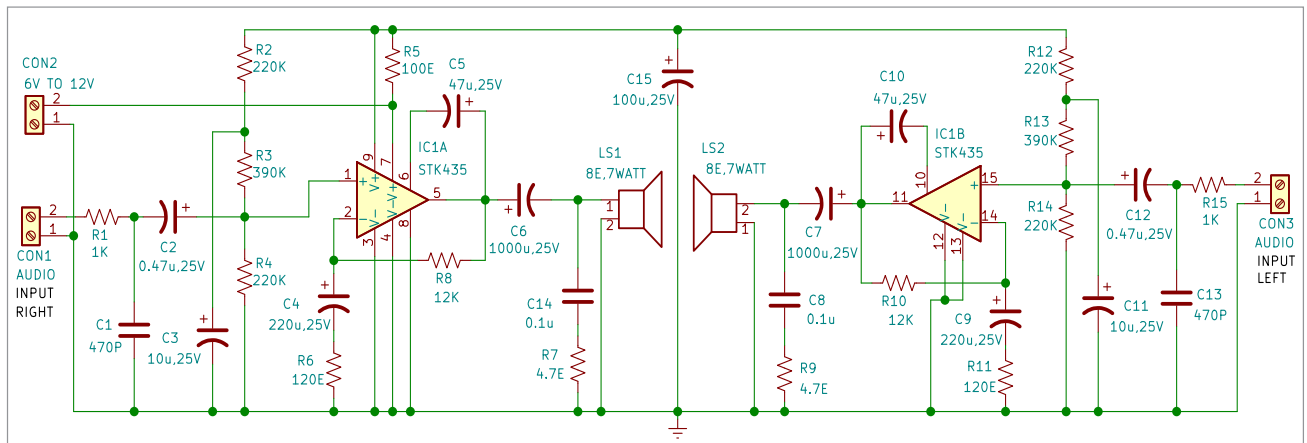


Fig. 2: Circuit diagram